

# Discrete Generators on Pseudotrees

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*Dedicated to the memory of Peter J. Nyikos.  
Without some of your final thoughts, none of this would be possible.*

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# Introduction

Previously known result:

## Baby Theorem

If  $T$  is a finitely branching tree, then  $T$  admits a discrete  $(0, 1)$ -generator whose members are all  $\{0, 1\}$ -valued.

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Ask any eighth grader.

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## Proof

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Joking. See my website:

<https://flemingchase.github.io/writings>.

# Questions

## An Interesting Question

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## An Actually Interesting Question

If a topological space is “tree-like,” what properties must it have to admit a binary generator?

## Answer

## Main Theorem

Let  $T$  be a Hausdorff, immediately branching, MUB-valent pseudotree. Then the following are equivalent:

- (i) There exists a homogeneous pair  $(\mathcal{M}, \mathcal{A})$  that finitely separates every point  $t \in T$ .
- (ii)  $T$  admits a binary generator.

# Pseudotrees

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## Example (Fake)

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## Example (Real)

$T = \{0\} \cup (\{0, 1\} \times (0, 1))$  with the ordering  $(i, r) < (j, s)$  iff  $i = j$  and  $r < s$  along with  $0 < (i, r)$  for all  $i \in \{0, 1\}$  and  $r \in (0, 1)$ .

# Goals

The goal of the following program is the following:

- (i) describe, by branching conditions only, how to recover tree structure from pseudotrees,
- (ii) determine what properties of those types of pseudotrees admit  $(0, 1)$ -generators, and
- (iii) give a concrete example of a (pseudo)tree that does not admit one.

# Discretely Branching Pseudotrees

## Definition: Valent sets

Let  $X$  be a poset and let  $F \subset X$  be an antichain. We say that  $F$  is *valent with respect to*  $X$  if each  $x \in X$  is comparable with a unique  $f \in F$ .

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## Useful Notation

For a point  $t$  in a pseudotree  $T$ ,  $t^> = \{t' \in T : t' > t\}$  and  $t^< = \{t' \in T : t' < t\}$ .

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## Definition: Valent at a Point

For a point  $t$  in a pseudotree  $T$  and an antichain  $F \subset t^>$ , we say that  $F$  is *valent at*  $t$  if it is valent with respect to  $t^>$ .

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## Definition: Valence of a Point

Let  $t$  be a point in a pseudotree  $T$ . The *valence at*  $t$  is defined to be

$$\lambda_t := \min\{|F| : F \text{ is a maximal antichain in } t^>\}.$$

# Discretely Branching Pseudotrees

## Definition: Discretely Branching Points

A point  $t$  is said to be *discretely branching* if there exists  $F \subset t^>$  that is valent at  $t$ . Otherwise,  $t$  is said to be *indiscretely branching*.

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## Trees are Discretely Branching

Let  $T$  be a tree. For any non-maximal point  $t \in T$ , let  $t^+$  be the set of immediate successors to  $t$ . Then  $t^+$  is valent with respect to  $t$ .

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This is pretty easy to see. If  $s > t$ , then, then there's some full branch of  $T$  that contains both  $s$  and  $t$ . Then there must be some immediate successor  $s'$  of  $t$  such that  $s'$  and  $s$  are comparable (possibly equal). Moreover,  $s'$  is unique.

Easy. So what does this matter?

# Discretely Branching Pseudotrees

## Proposition: Valent Sets Have Equal Cardinality

Let  $t$  be a discretely branching point in a pseudotree  $T$ . If  $F$  is valent at  $t$ , then  $|F| = \lambda_t$ .

## Definition: Wedges Above a Point

Let  $t$  be a non-maximal, discretely branching point in a pseudotree  $T$ , and let  $\mathcal{F}_t := \{F \subset t^> : F \text{ is valent at } t\}$ . For  $F \in \mathcal{F}_t$  and  $s \in F$ , define the *wedge from  $t$  toward  $s$*  to be

$$V_t(s) := \{t' \in t^> : t' \text{ is comparable with } s\}.$$

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## Proposition: Canonical Wedges above $t$

Let  $t$  be a non-maximal, discretely branching point in a pseudotree  $T$ . Then there exists a partition

$$\mathcal{W}_t = \{V_\lambda \subset t^> : \lambda < \lambda_t\}$$

of  $t^>$  such that for every valent set  $F$  at  $t$ , there is a bijection  $\tau : F \rightarrow \mathcal{W}_t$  such that  $\tau(s) = V_t(s)$ .

# Discretely Branching Pseudotrees

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## Proposition: Tree Partition Structure

Let  $T$  be a tree. For all  $t \in T$ , let  $t^+$  be the set of *immediate* successors of  $t$ . Then  $\{s^\geq : s \in t^+\}$  where  $s^\geq = \{t' \in T : t' \geq s\}$  partitions  $t^>$ .

# Immediately Branching Pseudotrees

## Definition: Immediately Branching Point

Let  $t$  be a non-maximal, discretely branching point in a pseudotree  $T$ . We say that  $t$  is *immediately branching* if, for every  $V_\lambda \in \mathcal{W}_t$ ,  $V_\lambda$  has a minimal element.

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We say that a discretely branching pseudotree  $T$  is *immediately branching* if every non-maximal point  $t \in T$  is immediately branching.

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## Definition: Immediately Branching Pseudotree

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## Proposition: Immediately Branching Tree-like Structure

Let  $t$  be a non-maximal, immediately branching point in a pseudotree  $T$ . For each  $\lambda < \lambda_t$ , let  $m_\lambda$  be the least element of  $V_\lambda$ . Then

$$t^+ = \{m_\lambda : \lambda < \lambda_t\}$$

and  $t^+$  is valent at  $t$ .

# MUB-valent Pseudotrees

## Definition: Dedekind Completeness

A poset  $X$  is *Dedekind complete* if every non-empty chain that is bounded above has a least upper bound. If a least upper bound  $c$  for a chain  $C$  exists,  $c$  is unique by definition.

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## Definition: Upper Bounds and Minimal Upper Bounds

Let  $T$  be a pseudotree, and let  $C$  be a non-empty bounded chain. We define the sets

$$\text{UB}(C) := \{t \in T : \forall c \in C, c \leq t\}$$

and

$$\text{MUB}(C) := \{m \in \text{UB}(C) : m \text{ is minimal in } \text{UB}(C)\}.$$

These are, respectively, the set of upper bounds of  $C$  and the set of minimal upper bounds of  $C$ .

# MUB-valent Pseudotrees

## Definition: MUB-valent Pseudotree

We say that  $T$  is *MUB-valent* if for every non-empty bounded chain  $C \subset T$ ,  $\text{MUB}(C)$  is valent with respect to  $\text{UB}(C)$ .

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Every tree is MUB-valent.

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If  $T$  is Dedekind complete, then  $T$  is MUB-valent. The converse is false.

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## Theorem: MUB-valent Characterization

Let  $T$  be an immediately branching pseudotree. Then  $T$  is not MUB-valent if and only if there exists a point  $t \in T$  such that  $\omega^*$  embeds into  $t^>$ .

# Recovering Trees from Pseudotrees

## Theorem: Cone Above a Point is a Tree

Let  $T$  be an immediately branching, MUB-valent pseudotree. Then, for all  $s, t \in T$  with  $s < t$ , the set  $(s, t]$  is well-ordered. Consequently,  $t^{\geq}$  is a tree.

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## Corollary

Let  $T$  be a pseudotree. Then  $T$  is immediately branching, MUB-valent, and has a unique least element if and only if  $T$  is a rooted tree.

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The preceding corollary summarizes our order-theoretic goals. The definition of a tree requires well-ordered predecessor sets, which is a global condition. For pseudotrees with a least element, immediate branching and MUB-valence recover this condition from only local branching and bounded chain properties.

# Hausdorff Equivalences

## Definition: Upper-limit Topology

Let  $X$  be a partially ordered set. Define

$$\mathcal{S} = \{x^> : x \in X\} \cup \{x^{\leq} : x \in X\}.$$

The topology generated by  $\mathcal{S}$  as a subbase is called the *upper-limit topology*.

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## Proposition: Upper-limit Local Base

Let  $T$  be a pseudotree endowed with the upper-limit topology. If  $t$  is non-minimal, then  $\{(s, t] : s < t\}$  is a local base at  $t$ . If  $t$  is minimal, then  $t$  is isolated.

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This topology is basically the half open interval topology. Moreover, it coincides with the usual topology on trees. Basic open sets look like half-open intervals of the form  $(s, t]$ .

# Hausdorff Equivalences

## Definition: Zero-Dimensional & Tychonoff

We adopt the convention that a space is *zero-dimensional* if it has a base of clopen sets, and that a space is *Tychonoff* if it is Hausdorff and completely regular.

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Completely regular just means that for any closed set  $A \subset X$  and a point  $x \notin A$ , there exists a continuous real-valued  $f : X \rightarrow \mathbb{R}$  such that  $f(x) = 1$  and  $f(A) = \{0\}$ .

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## Theorem: Hausdorff Equivalences

Let  $T$  be a MUB-valent pseudotree. Then the following are equivalent for  $T$ :

- (i) Hausdorff,
- (ii) Dedekind complete,
- (iii) zero-dimensional,
- (iv) Tychonoff.

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From here on out, every space considered is a MUB-valent Hausdorff Pseudotree.

# Limit Structure

## Definition: Relative Order Type

Let  $t$  be a non-minimal point and  $s < t$ . Since  $s^{\geq}$  is a tree, the interval  $t^{<} \cap s^{\geq} = [s, t)$  is well-ordered. Define the *relative order type of  $t$  with respect to  $s$*  to be

$$h_s(t) := \text{otp}([s, t)).$$

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## Definition: Relative Height

Let  $t$  be a non-minimal point of  $T$ .

- (i) We say  $t$  has *relative limit height* if there exists  $s < t$  such that  $h_s(t)$  is a non-zero limit ordinal.
- (ii) We say  $t$  has *relative successor height* if there exists  $s < t$  such that  $h_s(t)$  is a successor ordinal.

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## Proposition: Relative Successor

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## Proposition: Relative Limit Height Independence

If a non-minimal point  $t$  has relative limit height with respect to some  $s \in T$  with  $s < t$ , then it has relative limit height with respect to every  $s' < t$ .

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## Proposition: Point Trichotomy

Every point of  $T$  is either

- (i) minimal,
- (ii) an immediate successor, or
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We now simply say non-minimal points are either *limit points* or *immediate successors* with no reference to their relative height.

# Limit Structure

## Definition: Cofinality

Let  $\alpha$  be an ordinal. For  $A \subset \alpha$ , we say that  $A$  is *cofinal* in  $\alpha$  if for every  $\beta < \alpha$ , there exists  $\gamma \in A$  with  $\beta \leq \gamma$ . The *cofinality* of an ordinal is the cardinal  $\text{cf}(\alpha) = \min\{|A| : A \text{ cofinal in } \alpha\}$ .

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## Proposition: Relative Cofinality Independence

Let  $t$  be a limit point and  $s, s' < t$ . Then  $\text{cf}(h_s(t)) = \text{cf}(h_{s'}(t))$ .

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Let  $t$  be a limit point and  $s, s' < t$ . Then  $\text{cf}(h_s(t)) = \text{cf}(h_{s'}(t))$ .

## Definition: Relative Cofinality

Let  $t$  be a limit point. We define the *relative cofinality* of  $t$ , denoted  $\text{cf}_T(t)$ , to be  $\text{cf}(h_s(t))$  for any  $s < t$ . If  $t$  is an immediate successor or is minimal, we set  $\text{cf}_T(t) = 1$ .

# Limit Structure

## Proposition: Limit Structure of MUB-valent Hausdorff Pseudotrees

If  $t$  is a limit point, then there exists a strictly increasing sequence  $\langle t_\mu : \mu < \text{cf}_T(t) \rangle$  with  $t_\mu < t$  for every  $\mu < \text{cf}_T(t)$  such that  $t_\mu \rightarrow t$ . If  $t$  is an immediate successor or is minimal, then  $t$  is isolated.

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## Definition: Limiting Maps

Let  $t \in T$ . A map  $\phi_t : \text{cf}_T(t) \rightarrow T$  is called *limiting* if

- (i)  $\phi_t$  is strictly increasing and cofinal in  $t^<$  when  $t$  is a limit point,
- (ii)  $\phi_t(0) = t^-$  when  $t$  is an immediate successor, or
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## Definition: Uniform Limiting Collection

A collection of maps  $\mathcal{M} = \{\phi_t : t \in T\}$  is *uniform limiting* if  $\phi_t$  is limiting for each  $t$ .

# Limit Structure

## Definition: Monotone Clopen Local Base

An indexed clopen local base  $\mathcal{A}_t = \{A_{\mu,t} : \mu < \text{cf}_T(t)\}$  at a point  $t$  is called *monotone* if

- (i)  $A_{\mu,t} \subset t^{\leq}$  for every  $\mu < \text{cf}_T(t)$ , and
- (ii)  $A_{\nu,t} \subsetneq A_{\mu,t}$  for every  $\mu < \nu < \text{cf}_T(t)$ .

A collection of clopen local bases  $\mathcal{A} = \{\mathcal{A}_t : t \in T\}$  is *monotone* if each  $\mathcal{A}_t$  is monotone.

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A collection of clopen local bases  $\mathcal{A} = \{\mathcal{A}_t : t \in T\}$  is *monotone* if each  $\mathcal{A}_t$  is monotone.

## Definition: Homogeneous Pair

Let  $\mathcal{M}$  be a uniform limiting collection, and let  $\mathcal{A}$  be a monotone collection of clopen local bases. We call the pair  $(\mathcal{M}, \mathcal{A})$  *homogeneous* if, for each limit point  $t$  and  $\mu < \nu < \text{cf}_T(t)$ , the following chain of inclusions holds:

$$A_{\nu,t} \subset (\phi_t(\nu), t] \subsetneq A_{\mu,t} \subset (\phi_t(\mu), t].$$

# Limit Structure

## Proposition: Same Point Separation

Let  $(\mathcal{M}, \mathcal{A})$  be a homogeneous pair, and let  $t$  be a limit point. For any fixed  $\mu < \text{cf}_T(t)$ , there exists a finite set  $E_{\mu,t} \in [T]^{<\omega}$  such that for all  $\nu < \text{cf}_T(t)$  with  $\nu \neq \mu$ ,

$$E_{\mu,t} \cap (A_{\mu,t} \Delta A_{\nu,t}) \neq \emptyset.$$

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$$E_{\mu,t} \cap (A_{\mu,t} \Delta A_{\nu,t}) \neq \emptyset.$$

The previous proposition essentially says the follow: if we are given a homogeneous pair on a pseudotree, then distinct neighborhoods of a limit point can always be detected to be distinct by finitely many points.

# Binary Generators

## Definition: Generator

Let  $X$  be a Tychonoff space. A collection  $\mathcal{G}$  of real-valued continuous functions on  $X$  is a *generator* for  $X$  if, for every  $x \in X$  and every closed set  $K \subset X$  such that  $x \notin K$ , there exists  $g \in \mathcal{G}$  such that  $g(x) \notin \overline{g(K)}$ .

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## Definition: $C_p(X)$ neighborhoods

Let  $C_p(X)$  denote the set of all continuous real-valued functions on a Tychonoff space  $X$ , endowed with the topology of pointwise convergence. Basic open neighborhoods of  $f \in C_p(X)$  are the sets of the form

$$U(f, F, \epsilon) = \{g \in C_p(X) : \forall x \in F, |f(x) - g(x)| < \epsilon\}$$

where  $\epsilon > 0$  and  $F \in [X]^{<\omega}$ .

## Definition: Discrete Generator

A generator  $\mathcal{G}$  is called *discrete* if it is a discrete subspace of  $C_p(X)$ .

# Binary Generators

## Definition: $(0, 1)$ -generator

A generator  $\mathcal{G}$  is a  $(0, 1)$ -generator if, for every  $x \in X$  and every closed set  $K \subset X$  such that  $x \notin K$ , there is  $g \in \mathcal{G}$  such that  $g(x) = 1$  and  $g(K) \subset \{0\}$ .

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## Definition: Binary Generators

A discrete  $(0, 1)$ -generator  $\mathcal{G}$  is called a *binary generator* if  $\text{ran}(g) \subset \{0, 1\}$  for every  $g \in \mathcal{G}$ .

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## Definition: Binary Generators

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## Proposition: $\{0, 1\}$ -valued Continuous Functions

A continuous real-valued function  $f : X \rightarrow \mathbb{R}$  with range contained in  $\{0, 1\}$  is of the form  $\chi_A$  for some clopen set  $A \subset X$ . Conversely,  $\chi_A$  is continuous whenever  $A$  is clopen.

# Characterization Theorem

## Definition: Finite Separation

Let  $(\mathcal{M}, \mathcal{A})$  be a homogeneous pair. The pair *finitely separates at  $t$*  if, for every  $\mu < \text{cf}_T(t)$ , there exists a finite set  $F_{\mu,t} \in [T]^{<\omega}$  such that, for every  $t' > t$  and every  $\nu < \text{cf}_T(t')$ , if  $t \in A_{\nu,t'}$ , then

$$(A_{\mu,t} \Delta A_{\nu,t'}) \cap F_{\mu,t} \neq \emptyset.$$

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$$(A_{\mu,t} \Delta A_{\nu,t'}) \cap F_{\mu,t} \neq \emptyset.$$

## Theorem: Characterization Theorem

Let  $T$  be a Hausdorff, immediately branching, MUB-valent pseudotree. Then the following are equivalent:

- (i) There exists a homogeneous pair  $(\mathcal{M}, \mathcal{A})$  that finitely separates every point  $t \in T$ .
- (ii)  $T$  admits a binary generator.

# Sufficiency

Proof: Sufficiency

Read the paper :)

# Necessity

Assume now that  $\mathcal{G}$  is a binary generator for an immediately branching, MUB-valent, Hausdorff pseudotree  $T$ .

## Lemma: Clopen Base

For each limit point  $t \in T$  and each strictly increasing cofinal sequence  $\langle t_\mu : \mu < \text{cf}_T(t) \rangle \subset t^<$ , the generator  $\mathcal{G}$  contains the characteristic functions of a clopen local base  $\mathcal{V}_t = \{B_{\mu,t} : \mu < \text{cf}_T(t)\}$  at  $t$  such that  $B_{\mu,t} \subset (t_\mu, t]$ .

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## Proof Outline

Fix a strictly increasing cofinal sequence  $\langle t_\mu : \mu < \text{cf}_T(t) \rangle$ . For each  $\mu < \text{cf}_T(t)$ , the interval  $(t_\mu, t]$  is clopen. Hence  $(t_\mu, t]^c$  is closed. Use the generator to obtain a clopen set  $B_{\mu,t}$  for each  $\mu < \text{cf}_T(t)$ . Check that it is indeed a base and  $B_{\mu,t} \subset (t_\mu, t]$ .

# Necessity

## Lemma: Monotone Clopen Local Base from Generator

In the setting of the previous lemma, there exists a cofinal strictly increasing sequence  $\langle i_\mu : \mu < \text{cf}_T(t) \rangle \subset \text{cf}_T(t)$ , a monotone clopen local base  $\mathcal{A}_t = \{B_{i_\mu, t} : \mu < \text{cf}_T(t)\}$ , and a limiting map  $\phi_t$ .

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## Proof

Let  $\kappa = \text{cf}_T(t)$ . Given a strictly increasing cofinal sequence  $\langle t_\mu : \mu < \kappa \rangle$ , let  $\mathcal{V}_t = \{B_{\mu, t} : \mu < \kappa\}$  be the clopen local base from the previous lemma. We construct a subsequence of indices  $\langle i_\mu : \mu < \kappa \rangle$  in  $\kappa$  recursively.

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For  $\mu = 0$ , set  $i_0 = 0$ . Assume  $i_\mu$  has been defined. Since  $B_{i_\mu, t}$  is an open neighborhood of  $t$ , there exists  $s < t$  such that  $(s, t] \subset B_{i_\mu, t}$ . As  $\langle t_\mu : \mu < \kappa \rangle$  is cofinal, there exists  $\eta < \kappa$  such that  $t_\eta > s$ . Hence,

$$(t_\eta, t] \subsetneq (s, t] \subset B_{i_\mu, t}.$$

# Necessity

## Proof cont.

Set  $i_{\mu+1} = \max\{i_\mu, \eta\} + 1$ . Note that  $i_{\mu+1} > i_\mu$ , and if  $i_\mu \geq \mu$ , then  $i_{\mu+1} \geq \mu + 1$ . The inclusion

$$(t_{i_{\mu+1}}, t] \subsetneq B_{i_\mu, t}$$

follows from  $(t_{i_{\mu+1}}, t] \subsetneq (t_\eta, t] \subset B_{i_\mu, t}$  since  $t_\eta < t_{i_{\mu+1}}$  because  $i_{\mu+1} > \eta$  and  $\langle t_\mu : \mu < \kappa \rangle$  is strictly increasing.

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Assume  $i_\mu$  has been defined for all  $\mu < \delta$ , where  $\delta < \kappa$  is a non-zero limit ordinal. Since  $\delta < \kappa$  and  $\kappa$  is a regular cardinal,  $\gamma := \sup\{i_\mu : \mu < \delta\} < \kappa$ . Set  $i_\delta = \max\{\gamma + 1, \delta\}$ . For each  $\mu < \delta$ ,  $i_\delta > i_\mu$  since  $i_\delta \geq \gamma + 1 > \gamma \geq i_\mu$ , and  $i_\delta \geq \delta$ .

# Necessity

## Proof cont.

Set  $i_{\mu+1} = \max\{i_\mu, \eta\} + 1$ . Note that  $i_{\mu+1} > i_\mu$ , and if  $i_\mu \geq \mu$ , then  $i_{\mu+1} \geq \mu + 1$ . The inclusion

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The sequence  $\langle i_\mu : \mu < \kappa \rangle$  is strictly increasing and satisfies  $i_\mu \geq \mu$ . Hence  $\langle i_\mu : \mu < \kappa \rangle$  is cofinal in  $\kappa$ .

# Necessity

## Proof cont.

Consider the collection  $\{B_{i_\mu, t} : \mu < \kappa\}$ . For every  $\mu < \nu < \kappa$ , the following chain of inclusions holds:

$$\begin{aligned} (t_{i_{\nu+1}}, t] \subsetneq B_{i_\nu, t} &\subset (t_{i_\nu}, t] \\ &\subset (t_{i_{\mu+1}}, t] \subsetneq B_{i_\mu, t} \subset (t_{i_\mu}, t]. \end{aligned} \tag{1}$$

The first and fourth inclusions follow from the construction, the second and fifth follow from the previous lemma, and the third follows from  $i_\mu < i_{\mu+1} \leq i_\nu$ . Hence,  $B_{i_\nu, t} \subsetneq B_{i_\mu, t}$ .

# Necessity

## Proof cont.

The collection is a local base at  $t$ . For any neighborhood  $(s, t]$ , there exists  $\mu < \kappa$  with  $s < t_\mu$ . Hence,

$$B_{i_\mu, t} \subset (t_{i_\mu}, t] \subset (t_\mu, t] \subset (s, t].$$

Define  $A_{\mu, t} = B_{i_\mu, t}$ . Then  $\mathcal{A}_t = \{A_{\mu, t} : \mu < \kappa\}$  is a monotone clopen local base at  $t$ .

Define  $\phi_t(\mu) = t_{i_\mu}$ . Since  $\langle i_\mu : \mu < \kappa \rangle$  is strictly increasing and cofinal in  $\kappa$ , and  $\langle t_\mu : \mu < \kappa \rangle$  is strictly increasing and cofinal in  $t^<$ ,  $\langle \phi_t(\mu) : \mu < \kappa \rangle$  is strictly increasing and cofinal in  $t^<$ . Hence  $\phi_t$  is limiting.

# Necessity

Lemma: Generator gives Homogeneous Pair

$\mathcal{G}$  yields a homogeneous pair  $(\mathcal{M}, \mathcal{A})$ .

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## Lemma: Generator gives Homogeneous Pair

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## Lemma: Homogeneous Pair Finitely Separates

The homogeneous pair  $(\mathcal{M}, \mathcal{A})$  determined by  $\mathcal{G}$  finitely separates at every point  $t \in T$ .

# Example

## Tree with no Binary Generator

There exists a rooted Hausdorff tree that does not admit a binary generator.

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## Proof

Let  $T = \omega^{\leq \omega}$  be the complete  $\omega$ -ary tree of height  $\omega + 1$  ordered by extension. For  $s, t \in T$ ,  $s \leq t$  if  $\text{dom}(s) \subset \text{dom}(t)$  and  $s(n) = t(n)$  for all  $n \in \text{dom}(s)$ . The limit points of  $T$  are functions  $t \in \omega^\omega$ , and every limit point is maximal. Every point  $t \in \omega^{<\omega}$  is isolated, and the empty function is the root of  $T$ . Moreover, since  $T$  is a tree with a root,  $T$  is MUB-valent and immediately branching. The tree  $T$  is also Dedekind complete.

# Example

## Proof cont.

Suppose, toward a contradiction, that there is a homogeneous pair  $(\mathcal{M}, \mathcal{A})$  that finitely separates every point  $t \in T$ . For isolated points, let  $F_{0,t}$  be the set guaranteed by finite separation.

# Example

## Proof cont.

Suppose, toward a contradiction, that there is a homogeneous pair  $(\mathcal{M}, \mathcal{A})$  that finitely separates every point  $t \in T$ . For isolated points, let  $F_{0,t}$  be the set guaranteed by finite separation.

Define  $\psi : \omega^\omega \rightarrow \omega^{<\omega}$  by

$$\psi(t) := \min(A_{0,t}).$$

This map is well-defined. For each  $t \in \omega^\omega$ ,  $A_{0,t} \subset t^{\leq}$ . Since  $t^{\leq}$  is well-ordered,  $A_{0,t}$  has a least element.

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Suppose, toward a contradiction, that there is a homogeneous pair  $(\mathcal{M}, \mathcal{A})$  that finitely separates every point  $t \in T$ . For isolated points, let  $F_{0,t}$  be the set guaranteed by finite separation.

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This map is well-defined. For each  $t \in \omega^\omega$ ,  $A_{0,t} \subset t^{\leq}$ . Since  $t^{\leq}$  is well-ordered,  $A_{0,t}$  has a least element.

Moreover,  $\psi(t) \in \omega^{<\omega}$ . Since  $\psi(t) \in A_{0,t}$ ,  $\psi(t)$  and  $t$  are comparable. Since  $t$  is maximal, we must have  $\psi(t) \leq t$ . If  $\psi(t) = t$ , then  $A_{0,t} = \{t\}$ . However, this is not an open neighborhood of  $t$ , so  $\psi(t) < t$ .

Let  $t \in \omega^{<\omega}$  with  $\text{dom}(t) = n$ . Define

$$E_t := \omega \setminus \{s(n) : s \in t^+, \exists f \in F_{0,t}, s \leq f\}.$$

# Example

Proof cont.

$|E_t| = \omega$  because each  $f \in F_{0,t}$  determines at most one  $s \in t^+$ , and hence at most one  $s(n) \in \omega$ .

# Example

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$|E_t| = \omega$  because each  $f \in F_{0,t}$  determines at most one  $s \in t^+$ , and hence at most one  $s(n) \in \omega$ .

We build an element  $x \in \omega^\omega$  recursively by defining a sequence of points  $p_n \in \omega^{n+1}$  for every  $n < \omega$ .

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For  $n = 0$ , choose  $k \in E_\emptyset$ . Define  $p_0 \in \omega^1$  by  $p_0(0) = k$ . Suppose  $p_n \in \omega^{n+1}$  has been defined. Choose  $k \in E_{p_n}$ . Then define  $p_{n+1} \in \omega^{n+2}$  by  $p_{n+1}(n+1) = k$  and  $p_{n+1}(i) = p_n(i)$  for every  $i < n+1$ .

# Example

## Proof cont.

$|E_t| = \omega$  because each  $f \in F_{0,t}$  determines at most one  $s \in t^+$ , and hence at most one  $s(n) \in \omega$ .

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Define the limit point  $x \in \omega^\omega$  by  $x(n) = p_n(n)$ . Consider the point  $\psi(x) \in \omega^{<\omega}$ . Let  $n = \text{dom}(\psi(x))$ . Since  $\psi(x) < x$ , we have  $\psi(x)(i) = x(i)$  for every  $i < n$ . Since  $\psi(x) \in A_{0,x}$ , finite separation guarantees a point

$$f \in F_{0,\psi(x)} \cap (A_{0,\psi(x)} \Delta A_{0,x}) = F_{0,\psi(x)} \cap (A_{0,x} \setminus \{\psi(x)\}).$$

# Example

## Proof cont.

Since  $\psi(x)$  is the minimum of  $A_{0,x}$  and  $A_{0,x}$  is well-ordered, we have  $\psi(x) < f$ . Thus there exists  $s \in \psi(x)^+$  such that  $\psi(x) < s \leq f \leq x$ . Hence  $x(n) = s(n)$ .

# Example

## Proof cont.

Since  $\psi(x)$  is the minimum of  $A_{0,x}$  and  $A_{0,x}$  is well-ordered, we have  $\psi(x) < f$ . Thus there exists  $s \in \psi(x)^+$  such that  $\psi(x) < s \leq f \leq x$ . Hence  $x(n) = s(n)$ . This contradicts the construction of  $x$ . The recursion chose  $x(n) \in E_{\psi(x)}$ , while the existence of  $s \leq f$  gives  $s(n) \notin E_{\psi(x)}$ .

Thus no homogeneous pair  $(\mathcal{M}, \mathcal{A})$  can finitely separate every point, and by the main theorem,  $T$  has no binary generator.

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